CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (CL/EC/ME/CE/EE/IT) Sem - I University Examination, May '11

MA-101 Engineering Mathematics-1

Date: 24.05.11, Tuesday Time: 01:30 p.m to 04:30 p.m Maximum Marks: 70

Instructions: (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION-I

Q-1 Test the convergence of the following series: **[11]**

(a) (i) $\frac{2}{3}+\frac{4}{9}+\frac{8}{27}......$ (ii) $\sum_{n=1}^{\infty}\frac{n!}{3^n}$ (iii) $\sum_{n=1}^{\infty}\cos\left(\frac{1}{n}\right)$ **[9]**

(b) Find Argument and modulus of $1+i\tan\alpha$. **[2]**

Q-2 Attempt the following: **[12]**

(i) Find the square roots of $-15-8i$. **[5]**

(ii) Using De Moivre's theorem, prove that **[4]**

(a) $\frac{(\cos 3\theta+i\sin 3\theta)^5(\cos 2\theta-i\sin 2\theta)^3}{(\cos 4\theta+i\sin 4\theta)^{-9}(\cos 5\theta+i\sin 5\theta)^9}=1$ (b) $\left(\sqrt{3}+i\right)^n+\left(\sqrt{3}-i\right)^n=2^{n+1}\cos\left(\frac{n\pi}{6}\right)$

(iii) Using ratio test discuss convergence of $\sum_{n=1}^{\infty}\frac{x^n}{2^n}$. **[3]**

OR

Q-2 Attempt the following: **[12]**

(i) Disscuss the convergence of alternating series $\sum_{n=0}^{\infty}\frac{(-1)^n}{n+3}$. **[5]**

(ii) Expand $\cos^6\theta$ in a series of cosines of multiple of θ. **[4]**

(iii) If $i^{i^{\cdot^{\cdot^{\infty}}}}=A+iB$ then prove that $\tan\left(\frac{\pi A}{2}\right)=\frac{B}{A}$. **[3]**

Q-3 Attempt the following: **[12]**

(i) Find the rank of A = $\begin{bmatrix} 4 & 2 & 3 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. **[3]**

(ii) Solve the following system of linear equations using Gauss elimination method. [5]

$x_1 + 2x_2 - x_3 = 3,$

$2x_1 + 5x_2 - 4x_3 = 5,$

$3x_1 + 4x_2 + 2x_3 = 12,$

(iii) Use Gauss Jordan method to find invers of the following matrix [4]

$$\begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}.$$

OR

Q-3 Attempt the following: **[12]**

(i) Solve the following system of linear equations: [5]

$x - y + 2z - 3w = 4,$

$4x + y + 2w = 7,$

$3y + z + 4w = -1,$

$y + 2w = -1.$

(ii) Find the rank & nullity of [4]

$$A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}.$$

(iii) Show that the following system of linear equation is inconsistent [3]

$x + y + z = 6,$

$x + 2y + 3z = 10,$

$x + z = -1.$

SECTION-II

Q-4 Attempt the following: **[11]**

(a) (i) Find y_n, if $y = \dfrac{x}{(x+1)^2}$. [4]

(ii) Find n^{th} order derivative of $y = \sin 2x \cos 4x$ [3]

(b) If $y = \log\left(x + \sqrt{x^2+1}\right)$, prove that $\left(x^2+1\right) y_{n+2} + (2n+1) x y_{n+1} + n^2 y_n = 0$ [4]

Q-5 Attempt the following: **[12]**

(a) (i) By using Taylor's theorem, Expand $f(x)=x^5-x^4+x^3-x^2+x-1$ in the powers of $(x-1)$ and hence find $f(0.99)$. [3]

(ii) Evaluate: $\lim_{x\to 0}\dfrac{xe^x-\log(1+x)}{x^2}$. [2]

(b) Attempt the following

(i) Verify Lagrange's Mean Value Theorem for the function $f(x)=\log x,\ x\in[1, e]$. [3]

(ii)State Rolle's theorem . [2]

(c) If $u=(1-2xy+y^2)^{-1/2}$, then show that $x\dfrac{\partial u}{\partial x}-y\dfrac{\partial u}{\partial y}=u^3y^2$. [2]

OR

Q-5 Attempt the following: **[12]**

(a) If $u=\log(\tan x+\tan y+\tan z)$, then show that $2x\dfrac{\partial u}{\partial x}+\sin 2y\dfrac{\partial u}{\partial y}+\sin 2z\dfrac{\partial u}{\partial z}=2$. [4]

(b) If $u=\sin^{-1}\left(\dfrac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, then prove that $x^2u_{xx}+2xyu_{xy}+y^2u_{yy}=-\dfrac{\sin 2u}{4\cos^3 u}$. [4]

(c) Attempt the following:

(i) If $u=x^2+y^2+z^2$, where $x=e^t,\ y=e^t\sin t,\ z=e^t\cos t$, then find $\dfrac{du}{dt}$. [2]

(ii) If $y\log(\cos x)=x\log(\sin y)$, then find $\dfrac{dy}{dx}$. [2]

Q-6 Attempt the following: **[12]**

(a) (i) find the Jacobian $\dfrac{\partial(u,v)}{\partial(x,y)}$ for $u=x+\dfrac{y^2}{x},\ v=\dfrac{y^2}{x}$. [3]

(ii) Find the percentage error in calculating the area of a rectangle when an error of 3% is made in measuring each of its sides. [2]

(b) Find the extreme values of $u=x^3+3xy^2-3x^2-3y^2+7$, if any. [4]

(c) Find the points of the surface $z^2=xy+1$ nearest to the origin. Also find that distance. [3]

OR

Q-6 Attempt the following: **[12]**

(a) Find the maximum value of $f(x,y,z)=x^2y^3z^4$ subject to the condition $x+y+z=5$ [4]

(b) Find approximate value of $[(0.98)^2+(2.01)^2+(1.94)^2]^{1/2}$. [4]

(c) A rectangular box open at top is to have a volume 108 cubic meters. Find the dimensions of the box if its total surface is minimum. [4]